MOH IRSHAD

M.Tech (Fuel & Energy Engineering), IIT (ISM) Dhanbad +91-77349-28875
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Education

Indian Institute of Technology (Indian School of Mines), Dhanbad 2025 – 2027 (Expected)
M.Tech in Fuel & Energy Engineering *Dhanbad, Jharkhand*

S.S. College of Engineering, Udaipur (RTU, Kota) 2020 – 2024
B.Tech in Mining Engineering *Udaipur, Rajasthan*

- CGPA: **9.41/10**
- Secured **4th Merit Rank** in Rajasthan Technical University (RTU), Kota.

Experience

Lakhampur Masonry Stone Mine (M/s Shri Ram Stone Crusher) Jun 2024 – Jun 2025
Graduate Engineer Trainee *Bharatpur, Rajasthan*

- Monitored daily production operations to ensure efficient process execution, equipment availability, and uninterrupted plant performance.
- Collected, analyzed, and maintained operational data to support process monitoring, productivity improvement, and engineering decision-making.
- Coordinated with cross-functional teams to optimize workflow, improve operational efficiency, and ensure compliance with safety and standard operating procedures (SOPs).
- Prepared technical reports, monitored equipment performance, and contributed to troubleshooting and continuous process improvement initiatives.

Projects

Comparative Analysis of Hydrogen Production from Municipal Solid Waste Using Aspen Plus Ongoing

- Currently developing Aspen Plus simulation models for hydrogen production from municipal solid waste (MSW) through (i) steam gasification, (ii) steam gasification integrated with water–gas shift, and (iii) steam methan reforming.
- Developing a harmonized simulation framework using a common feedstock, thermodynamic property method, and system boundary for comparative process evaluation.
- Evaluating hydrogen yield, syngas composition, energy efficiency, and exergy efficiency to compare thermochemical and biochemical conversion pathways.
- Performing sensitivity analysis to investigate the effects of gasification temperature, steam-to-feed ratio, and other operating parameters on hydrogen production performance.
- Validating Aspen Plus simulation results with published literature and preparing the study for research publication.

Technical Skills

- **Energy & Process Engineering:** Fuel Technology, Thermochemical Conversion, Biofuels, Green Hydrogen, Renewable Energy Systems
- **Process Simulation:** Aspen Plus, Process Modeling, Material & Energy Balance, Process Flow Diagram (PFD), Sensitivity Analysis
- **Laboratory Techniques:** Gas Chromatography (GC), Thermogravimetric Analysis (TGA), Batch Reactor Operation, Catalyst Preparation, Proximate & Ultimate Analysis
- **Programming & Software:** Aspen Plus, Python, MATLAB, MS Excel, AutoCAD
- **Research & Analysis:** Literature Review, Process Optimization, Data Analysis, Technical Writing, Scientific Documentation
- **Professional Skills:** Problem Solving, Team Collaboration, Technical Communication, Presentation Skills

Achievements

- **GATE 2025 Qualified** with a Score of **356** and All India Rank (**AIR 1133**).
- **Secured 4th Merit Rank** in B.Tech Mining Engineering under Rajasthan Technical University (RTU), Kota.
- Recipient of the **Certificate of Excellence**, **Certificate of Honour**, and **Certificate of Appreciation** for outstanding academic performance and technical excellence.

Leadership & Engagement

- Collaborated with multidisciplinary teams during industrial training and research projects, strengthening teamwork, technical communication, and problem-solving skills.
- Actively engaged in technical seminars, research discussions, and professional networking with academia and industry to enhance knowledge in fuel, energy, and process engineering.